

# Project Proposal

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Interior Layout Intelligent Planning System

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## Theme

Apply search, planning, constraint satisfaction, and reinforcement learning to automate indoor furniture layout and navigation.

## Course Relevance

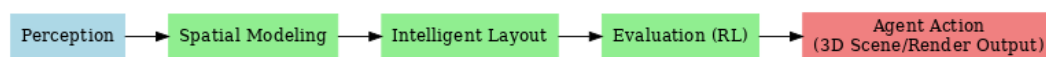
- Search & Planning → A\* path planning
- State Modeling → Room/furniture grids
- Constraint Satisfaction → Collision checks & corridor rules
- Reinforcement Learning → Q-learning navigation
- Visualization → 2D demo

## System Structure

The system has 3 layers and 5 modules:

1. Perception → Extract room and furniture info from images or synthetic data (YOLO or manual JSON).
2. Spatial Modeling → Build occupancy grids and clearance maps for state representation.
3. Intelligent Layout → Apply A\* and spatial constraints to generate/validate furniture layouts.
4. Evaluation (RL) → Train a Q-learning agent for navigation and measure success rate.
5. Agent Action → Output results as 2D visualization, extendable to 3D scenes or HD renders.

The diagram below illustrates the module flow:



## Request

Please confirm if this plan fits the course requirements and provide any advice.

Best regards,  
Qiyun Ge, Yuping Yan